NS-P02: Implementation and Verification Report

Executive summary

This project delivers an offline educational network-policy demonstration and a Cisco router/switch build kit. Four address zones protect the intended IT/OT boundary. Policy decisions, PAT mappings and simplified reply matching are implemented and tested. The attached Word report serves as a design source, not evidence that this implementation ran in Cisco Packet Tracer.

Workflow delivered

Open dashboard.html locally or serve web/index.html with Python. Use Live lab for custom traffic, Validation & evidence for the 20-check suite and JSON export, and Report coverage for the requirement audit. No additional Mac software is required for the browser workflow.

Policy decisions

Corporate and Admin may initiate HTTPS and approved DNS. The jump host 192.168.30.10 may initiate SSH to internal systems and RDP to SCADA 192.168.20.10. Corporate initiations to OT/Admin, OT Internet access, unsolicited Internet traffic, and routed Telnet are denied. Lab-only ping to 198.51.100.10 is a documented diagnostic exception. Corporate/Admin have PAT; OT is intentionally excluded.

Verification

Twenty browser checks passed, including scenario decisions, PAT uniqueness, matching replies and a wrong-source denial. Browser automation also verified presets, input errors, export, reset and responsive layout. Sixteen Python test methods passed. The ACL subset evaluator checked 5,334 Corporate boundary cases and 253 Admin source cases. The underlying IOS command parser, device forwarding, and actual TLS/SSH/RDP were not executed.

Stateful comparison

The browser and Python models match reverse flow tuples to simulated sessions. They do not validate TCP sequence numbers or implement FIN/RST teardown or idle expiration. The Cisco configuration uses stateless established checks, which accept ACK/RST flags even without a session table. The static ACL test demonstrates this weaker guarantee.

Read-only reporting demonstration

A separate Python standard-library SQLite demo passed 13 reporting checks: the dedicated-host policy tuple, allowed reads, rejected writes/schema changes/overrides and unchanged data. SQLite mode=ro, query_only and a restricted authorizer enforce read-only access on this connection. The assumed reporting host is 192.168.10.60 and the proposed service is 192.168.20.10 TCP 8443. The original strict-deny dashboard does not enable this exception. No HTTPS reporting listener, real cross-VLAN flow or industrial database synchronization is claimed. Local administrators can still open a separate writable connection.

Report corrections

Do not claim that segmentation eliminates ransomware or that NAT completely obscures topology. This is not an air gap. Gateway filters do not inspect same-VLAN flows. Say traffic was dropped

rather than connection refused unless a reject was observed. Do not reuse the source report's claimed 100% Cisco compliance as evidence for this implementation.

Completion status

The no-install browser demonstration and supporting materials are complete. The original Packet Tracer execution requirement is not complete. No .pkt topology, Cisco packet captures or real application handshakes are claimed. Production controls, including industrial DPI, endpoint hardening, USB restrictions and hardware data diodes, remain recommendations.